

OpenTDP "FIN TDP" Specification

Technical Data Package — Field Reference & File Format

Version	v20260919a · Alpha Release · 2026-09-19
Producer	V. Viljanen
Contact	fintdp@outlook.com
License	GNU General Public License v3 (GPLv3)
Fields	179 fields defined across 8 tabs
Identifier	OpenTDP "FIN TDP"
GitHub	https://github.com/aurismo/OpenTDP
Website	https://opentdp.eu
XML Schema	TDP_Schema_2026-09-19.xsd (bundled with the application)
Generated	2026-09-19

This document was generated by Claude.ai (Anthropic) based on the OpenTDP Package Editor project's source code, and reviewed for accuracy against that code before publication.

This is an Alpha version of both the OpenTDP Package Editor application and the OpenTDP "FIN TDP" specification. Features and field definitions may change in future releases.

About OpenTDP "FIN TDP"

The Technical Data Package (TDP) specification — released as OpenTDP "FIN TDP" — is designed as an open language for distributed and advanced manufacturing. Its goal is to enable interoperability and provide a standardised way to exchange manufacturing information across organisations, tools and platforms, regardless of the underlying software or production technology. The specification is open and freely implementable.

Source and issue tracker: <https://github.com/aurismo/OpenTDP>

Project website: <https://opentdp.eu>

This revision documents changes made to the OpenTDP Package Editor application since the 2026-09-06 release: the TDP.json mirror format (Section 1.6), the revised package-checksum algorithm and its History/ audit trail (Sections 1.4-1.5), the Manufacturing/ attachment folder layout (Section 1.2), and the new Sequence Letter field (Section 4). See the change note at the end of Section 1 for a complete list.

1. TDP File Format

1.1 Overview

A TDP package is stored as a single .tdp file. Internally it is a standard ZIP archive (RFC 1950 / DEFLATE compression) that bundles a structured XML manifest, a field-ID-keyed JSON mirror of that manifest, and all referenced binary assets and audit records — geometry files, build files, images, step attachments and per-save checksum logs. Renaming a .tdp file to .zip and opening it with any archive manager reveals its full contents.

1.2 Archive Structure

The following directory layout is used inside every .tdp archive:

Path inside archive	Contents
TDP.xml	Main XML document — all metadata, versioning, quality data, manufacturing options, steps and field values.
TDP.json	Machine-readable mirror of TDP.xml. Every value is keyed "d<fieldId>" instead of a tag name — see Section 1.6.
Geometry/	As-designed mesh models. Accepted formats: STL, 3MF. File names match GeometryFile/FileName in XML.
BuildFiles/	AM build files (G-code, AMF, 3MF, MBF, CLS, ...). File names match BuildFile/FileName in XML.
Optional/CAD/	Additional CAD or reference models (STL, STEP, IGES, 3MF). Not used for geometry validation — informational only.
Optional/Images/	Reference images (PNG, JPG, BMP, TIFF). Used for visual documentation of the part or process.
Manufacturing/<OptionID>/<StepID>/	Step-level attachments: one folder per Process Option and, within it, one folder per Process Step, mirroring the Manufacturing tab's own structure. Replaces the flat Attachments/ layout used before 2026-09-18 — see Section 5 for a worked example.
History/<ChangeID>.txt	One plain-text audit record per save, named after that save's Change ID (Section 1.5). Lists every field that went into that save's package checksum, in the order used, plus the OS account that was logged in at save time. Entries accumulate permanently; none are ever removed.

In addition to the archive itself, the application writes a loose checksum_debug.txt file next to the saved .tdp file on disk (not inside the archive) after every save — a copy of the most recent History/ entry, for quick inspection without unzipping anything.

1.3 TDP.xml Document Structure

The XML document has a single root element <TDP> with a version attribute indicating the schema version. Almost every field carries a fieldId attribute that corresponds to the field index number in this specification, enabling automated processing and round-trip validation; the small number of implementation-only values that do not (an internal Process Option identifier, and the per-save audit checksum on each Change History entry) are documented in Appendix A rather than the numbered Field Index. The top-level structure is:

A complete XML Schema (XSD) for TDP.xml is generated directly from this application's source code and ships alongside it as TDP_Schema_2026-09-19.xsd, opened from the application's Schema button next to Specification. Any TDP.xml written by a conforming save operation validates against it.

Element	Contains
<TDP version="1.0">	Root element
<Metadata>	Item ID, name, contact, licensing, lifecycle status, criticality level, classification
<Version>	Unique ID (UUID), revision stamp, package SHA-256 checksum, encryption settings, change history log
<Quality>	Traceability rules, risk classification (tech + safety), identified risks, approval and audit trail
<Geometry>	One <GeometryFile> per mesh model with file name, format, size, SHA-256 checksum and bounding box
<ManufacturingOverview>	General instructions, manufacturing method, primary material, applicable standards
<ProcessOptions>	One <ProcessOption> per manufacturing route, each containing ordered <Steps> with full process data

Element	Contains
<BuildFiles>	One <BuildFile> per AM build file with printer, material and certification sub-elements
<PackingAndSafety>	Packing, handling, transport and safety information
<OptionalData>	Additional CAD models and reference images

1.4 Integrity and Checksums

Every binary asset (geometry file, build file, optional file, step attachment) carries a SHA-256 checksum stored as a hex string in its corresponding XML element. Checksums are recomputed automatically on each save — except when the referenced source file is temporarily unreachable, in which case the previously computed value is kept rather than cleared, so a transient path issue never looks like the file's content changed. Any consumer can use these checksums to verify that the assets have not been modified since the package was created.

A package-level checksum is stored in Version/Checksum (11040). It is computed as the SHA-256 of a canonical string built by walking every fieldId-tagged value in TDP.xml, ordering them numerically by fieldId (and by document order for repeated fields, e.g. multiple GeometryFile records), and concatenating them with fixed field/entry separators. The following are deliberately excluded from this calculation, because they change on every save regardless of whether any real content changed: Metadata/ModifiedDate (10070), and the ChangeID (11100), Timestamp (11110), Author (11120) and Description (11130) of each Version/ChangeHistory/ChangeEntry. Version/Checksum itself (11040) is excluded as it would otherwise be self-referential. Everything else — including every individual asset's own SHA-256 — is included, so the package checksum still changes whenever any real content or referenced file changes, while staying identical across consecutive saves where nothing meaningful changed.

For full transparency, every field that contributed to a given save's checksum, in the exact order concatenated, is written to History/<ChangeID>.txt inside the archive and to the loose checksum_debug.txt next to the saved file (Section 1.2).

1.5 Change History

Every save operation appends a ChangeEntry record to Version/ChangeHistory. Each entry contains a unique change ID (format: CH-XXXXXXXX), a UTC timestamp, an optional author name, and a description — these four values are the ones defined at fieldId level (11100–11130). Each entry additionally carries a Checksum value recording the package content-checksum (Section 1.4) at the moment that entry was written. Because this value deliberately carries no fieldId, it never enters the very calculation it records, and it lets any reader see at a glance whether two consecutive saves actually changed anything: identical Checksum values across consecutive entries mean no tracked content changed between those saves. The history is append-only and is never modified by the application, providing a lightweight audit trail of all save events. The full field-by-field breakdown behind each entry's checksum is preserved permanently inside the archive as History/<ChangeID>.txt (Section 1.2), and the application's Version tab can open that file directly by double-clicking the corresponding Change History row.

1.6 TDP.json (Field-ID Keyed JSON Mirror)

Alongside TDP.xml, every .tdp archive also contains TDP.json: the same data, restructured so every key is the letter "d" followed by the field's numeric ID (e.g. "d10020") — never a tag name or other plain-language identifier. TDP.json is generated directly from TDP.xml, so the two files can never disagree about which value belongs to which field.

Repeating groups (multiple Geometry files, Process Options, Steps, Attachments, Parameters, Change History entries, ...) become JSON arrays. Where the group's XML container has no fieldId of its own, a synthetic, non-spec numeric id is used purely as a stable array key — Appendix A lists every such identifier. The same treatment is used for the handful of implementation-only leaf values that likewise carry no fieldId in TDP.xml (Process Option's internal identifier; each Change History entry's audit Checksum). Parameter is a special case: fieldId 15900 tags the parameter's name and 15910 its value, represented as {"d15900": name, "d15910": value} per instance.

If a fieldId-less element with real (non-empty) content is ever encountered without a synthetic id mapped for it, the JSON generator raises an error rather than silently omitting data — keeping Appendix A a complete and

trustworthy list.

1.7 Lifecycle and Criticality

Two orthogonal axes classify every TDP package:

Axis	Values	Determines
Lifecycle	Draft · In Review · Approved · Released · Deprecated	Which fields are required to save/advance
Criticality	Non-critical · Low · Medium · High · Very High	Derived from max(Technical Risk, Safety Risk)

The intersection of lifecycle and criticality determines the exact set of required fields at any point in time. See Section 3 (Lifecycle × Criticality Matrix) for the full mapping.

1.8 Changes Since 2026-09-06

- Root <TDP> now always carries the version="1.0" attribute.
- Element order in TDP.xml matches this document exactly (Quality now precedes Geometry, as always specified).
- Duration/Hours, Duration/Minutes, Standards/Standard, Inputs/Input, Outputs/Output and BuildFileRefs/Ref now all carry their documented fieldId attributes.
- Package checksum algorithm changed from a JSON snapshot to the fieldId-ordered canonical string described in Section 1.4, and now excludes ModifiedDate and Change History's own bookkeeping fields so it stays stable across saves with no real content change.
- Each Change History entry now records the package checksum at that moment (Section 1.5), and a full audit trail is kept in History/ inside the archive (Section 1.2).
- Step-level attachments moved from a flat Attachments/<StepID>_<file> layout to Manufacturing/<OptionID>/<StepID>/<file> (Section 1.2, Section 5).
- New TDP.json mirror of TDP.xml (Section 1.6, Appendix A).
- New field 15025 Sequence Letter, distinguishing Process Steps that share a Sequence Number because they can be performed simultaneously (Section 4).
- Validation now enforces Unique ID (always) and Approver / Approval Conditions / Evidence References (Approved or Released, Low+) as described in Section 2 — these requirements were already documented but are now actually checked by the application.

2. Validation Gate Summary

Lifecycle / Condition	Required fields
Draft (all criticality levels)	Item ID · Name · Contact · Licensing · Unique ID · Lifecycle Status
In Review+ (Low+)	+ Geometry (STL or 3MF) + Identified Risks
In Review+ (Non-critical)	+ Geometry (STL or 3MF) [No Identified Risks required]
Approved / Released (all)	+ Revision Timestamp + Approval Status ≠ Pending
Approved / Released (Low+)	+ Approver · Approval Conditions · Evidence References
Approved / Released (Medium+)	+ Overview: Description · Manufacturing Method · Material
Approved / Released (High+)	+ ≥1 Process Option + ≥1 Process Step
Approved / Released (Very High+)	+ ≥1 Inspection Method in any Process Step
Non-critical special rules	In Review not available. Approval Status auto-set to Approved. Audit trail optional.

3. Lifecycle × Criticality Matrix

	Draft	In Review	Approved	Released	Deprecated
Non-critical	Item ID + Name Contact + Licensing	— N/A —	+ Geometry + Revision + Status = Approved	Same as Approved	Read-only
Low	Item ID + Name Contact + Licensing	+ Geometry + Identified Risks	+ Revision + Approval fields	Same as Approved	Read-only
Medium	Item ID + Name Contact + Licensing	+ Geometry + Identified Risks	+ Revision + Approval + Overview	Same as Approved	Read-only
High	Item ID + Name Contact + Licensing	+ Geometry + Identified Risks	+ Revision + Approval + Overview + ≥1 Option + ≥1 Step	Same as Approved	Read-only
Very High	Item ID + Name Contact + Licensing	+ Geometry + Identified Risks	+ Revision + Approval + Overview + ≥1 Option + ≥1 Step + ≥1 Inspection Method	Same as Approved	Read-only

4. Field Index

179 fields across 8 tabs. See Appendix A for a small number of implementation-only identifiers that exist in TDP.xml / TDP.json but are not part of this numbered index.

TAB: Metadata

ID	Title	XML Path	Type	N C	Lo	M e	Hi	V H	Notes
10010	Item ID (TDP ID)	Metadata/TDP_ID	Text	X	X	X	X	X	Required always — any lifecycle, any criticality. Primary persistent identifier. Max 100 chars.
10020	Additional IDs	Metadata/AdditionalIDs	Text						Optional. Other identifiers e.g. NSN. Max 200 chars.
10030	Item Name	Metadata/Name	Text	X	X	X	X	X	Required always — any lifecycle, any criticality. Max 200 chars.
10040	Description	Metadata/Description	Text (long)						Optional. Multi-line.
10050	Revision	Metadata/Version	Text						Optional free text e.g. 'A' / '1.0'. Max 50 chars.
10060	Created Date	Metadata/CreatedDate	Date						Optional. Format YYYY-MM-DD.
10070	Modified Date	Metadata/ModifiedDate	Date						Automatic. Updated on save. Read-only.
10080	Lifecycle Status	Metadata/LifecycleStatus	Text (select)	X	X	X	X	X	Required — drives all validation gates. Values: Draft In Review Approved Released Deprecated. Non-critical: In Review is not offered (skips straight to Approved) — see Section 2.
10090	Author	Metadata/Author	Text						Optional. Max 100 chars.
10100	Organization	Metadata/Organization	Text						Optional. Max 200 chars.
10110	Contact	Metadata/Contact	Text	X	X	X	X	X	Required always — establishes IP accountability from draft stage. Max 200 chars.
10120	Licensing	Metadata/Licensing	Text	X	X	X	X	X	Required always — establishes IP rights from draft stage. Max 200 chars.
10130	Information Classification	Metadata/InformationClassification	Text (select)						Optional. Values: Internal Public Unclassified Restricted Confidential Secret Top Secret. Default: Internal.
10140	Confidentiality Note	Metadata/ConfidentialityLevel	Text						Optional. Max 100 chars.
10150	Criticality Level (computed)	Metadata/CriticalityLevel	Text (select)						Automatic — derived from Quality tab risk scores. Read-only. Values: Non-critical Low Medium High Very High.

TAB: Version Control

ID	Title	XML Path	Type	N C	Lo	M e	Hi	V H	Notes
11010	Unique ID (UUID)	Version/UniqueID	Text (UUID)	X	X	X	X	X	Required always — package permanent identity. RFC 4122 UUID. Auto-generated. Max 36 chars.
11020	Assembly ID	Version/AssemblyID	Text						Optional. Reference to parent assembly. Max 100 chars.
11030	Revision Timestamp	Version/Revision	Text				X	X	Required when lifecycle ≥ Approved. Recommended format: YYYYMMDDHHmmss. Max 20 chars.
11040	Package Checksum (SHA-256)	Version/Checksum	Text (hash)						Automatic. 64 hex chars. Computed on save. Read-only.
11050	Signature	Version/Signature	Text						Optional. Max 1000 chars.
11060	Encryption Enabled	Version/EncryptionEnabled	Boolean						Optional. true false. Default: false.
11070	Encryption Algorithm	Version/EncryptionAlgorithm	Text (select)						Required if encryption = true. Values: (blank) AES-256 Fernet RSA2048.

ID	Title	XML Path	Type	N C	Lo	M e	Hi	V H	Notes
Change History — repeating record (one entry per save)									
11100	Change ID	Version/ChangeHistory/ChangeEntry/ChangeID	Text						Automatic. Format: CH-XXXXXXX.
11110	Timestamp	Version/ChangeHistory/ChangeEntry/Timestamp	DateTime						Automatic. Format: YYYY-MM-DD HH:MM:SS.
11120	Author	Version/ChangeHistory/ChangeEntry/Author	Text						Automatic / manual. Max 100 chars.
11130	Description	Version/ChangeHistory/ChangeEntry/Description	Text						Automatic / manual.

Implementation note: each ChangeEntry additionally carries a Checksum value (the package content-checksum at that save, Section 1.5). It carries no fieldId in TDP.xml and is not part of this numbered index; in TDP.json it appears under the synthetic key d11095 (Appendix A).

TAB: Quality Assurance

ID	Title	XML Path	Type	N C	Lo	M e	Hi	V H	Notes
Traceability and Identification Rules									
18010	Serial Number Format	Quality/Traceability/SerialNumberFormat	Text						Optional. Max 200 chars.
18020	Batch ID Format	Quality/Traceability/BatchIDFormat	Text						Optional. Max 200 chars.
18030	Material Lot Format	Quality/Traceability/MaterialLotFormat	Text						Optional. Max 200 chars.
18040	Labeling Requirements	Quality/Traceability/LabelingRequirements	Text						Optional. Max 500 chars.
18050	Traceability Level	Quality/Traceability/TraceabilityLevel	Text (select)						Optional. Granularity required: Batch Lot Unit Component.
Risk Classification → Determines Criticality Level									
18100	Technical Performance Risk	Quality/Risks/TechnicalPerformance	Text (select)	X	X	X	X	X	Required always — determines criticality. Values, least to most severe: Negligible Technical Impact Low Technical Impact Moderate Technical Impact High Technical Impact Critical Technical Impact.
18110	Safety Criticality Risk	Quality/Risks/SafetyCriticality	Text (select)	X	X	X	X	X	Required always — determines criticality. Criticality = the more severe of this and Technical Performance Risk. Values, least to most severe: Minimal Safety Concern Low Safety Concern Moderate Safety Concern High Safety Concern Critical Safety Hazard.
18120	Identified Risks (free text)	Quality/Risks/IdentifiedRisks	Text (long)		X	X	X	X	Required when lifecycle ≥ In Review and criticality ≥ Low. Multi-line.
General Process Quality									
18200	Process Quality Notes	Quality/ProcessQualityNotes	Text (long)						Optional. Multi-line.
Approval and Audit Trail									
18300	Approver	Quality/Approval/Approver	Text			X	X	X	Required when lifecycle = Approved or Released (Low+ only — Non-critical: optional). Max 100 chars.
18310	Approval Status	Quality/Approval/Status	Text (select)			X	X	X	Required when lifecycle = Approved or Released. Values: Pending Approved Rejected. Default: Pending. Non-critical: automatically set to Approved — see Section 2.
18320	Approval Timestamp	Quality/Approval/Timestamp	DateTime						Optional. Format: YYYY-MM-DD HH:MM:SS.

ID	Title	XML Path	Type	N C	Lo	M e	Hi	V H	Notes
18330	Approval Conditions	Quality/Approval/Conditions	Text (long)			X	X	X	Required when lifecycle = Approved or Released (Low+ only). What must be submitted/demonstrated. Multi-line.
18340	Evidence References	Quality/Approval/EvidenceRefs	Text			X	X	X	Required when lifecycle = Approved or Released (Low+ only). References to submitted evidence. Max 500 chars.

TAB: Geometry

ID	Title	XML Path	Type	N C	Lo	M e	Hi	V H	Notes
Geometry Files — Repeating Record (one per as-designed mesh model)									
12010	Geometry ID	Geometry/GeometryFile/id	Text						Automatic. Format: Gnnn (e.g. G001). Read-only.
12020	File Name	Geometry/GeometryFile/FileName	Text						Automatic from file selection. Read-only.
12030	File Format	Geometry/GeometryFile/FileFormat	Text						Automatic. Accepted formats: STL 3MF only (open triangle-mesh). STEP, IGES, OBJ not accepted. Max 10 chars.
12040	File Size (bytes)	Geometry/GeometryFile/FileSize	Text (number)						Automatic. Warning (grey) if < 100 bytes. Read-only.
12050	SHA-256 Checksum	Geometry/GeometryFile/Checksum	Text (hash)						Automatic. 64 hex chars. Read-only.
12060	Dimensions	Geometry/GeometryFile/Dimensions	Text						Automatic. Trimesh computes bounding box. Format: 'X × Y × Z mm'. Red warning if cannot be computed. Read-only.
12070	Preview	Geometry/GeometryFile/Preview	Boolean						User selects. At most one at a time (radio-button). true false.
Package-level Geometry Requirement									
12001	≥1 geometry file (as-designed mesh model)	(package-level requirement)	Requirement		X	X	X	X	Not a single field — package-level requirement. Accepted formats: STL or 3MF. Scanned or reverse-engineered reference models belong in Additional Data (Optional/CAD), not here.

TAB: Build Data

ID	Title	XML Path	Type	N C	Lo	M e	Hi	V H	Notes
AM Build Files — Repeating Record (one per file)									
16010	Build File ID	BuildFiles/BuildFile/id	Text						Automatic. Format: BF-nnn. Read-only.
16020	File Name	BuildFiles/BuildFile/FileName	Text						Automatic. Read-only.
16030	File Format	BuildFiles/BuildFile/FileFormat	Text						Automatic. Supported: GCODE AMF 3MF MBF CLS. Max 10 chars.
16040	File Size (bytes)	BuildFiles/BuildFile/FileSize	Text (number)						Automatic. Read-only.
16050	File Type	BuildFiles/BuildFile/FileType	Text						User entry. Default: 'AM Build File'. Max 100 chars.
16060	SHA-256 Checksum	BuildFiles/BuildFile/Checksum	Text (hash)						Automatic. 64 hex chars. Read-only.
Machine Information									
16100	Printer Manufacturer	BuildFiles/BuildFile/Printer/Manufacturer	Text						Completeness: grey row if missing. Max 100 chars.
16110	Printer Model	BuildFiles/BuildFile/Printer/Model	Text						Completeness: grey row if missing. Max 100 chars.
16120	Printer Serial Number	BuildFiles/BuildFile/Printer/SerialNumber	Text						Optional. Max 100 chars.
Material Information									
16200	Material Name	BuildFiles/BuildFile/Material/Name	Text						Completeness: grey row if missing. Max 100 chars.
16210	Material Grade	BuildFiles/BuildFile/Material/Grade	Text						Optional. Max 100 chars.
16220	Material Batch Number	BuildFiles/BuildFile/Material/BatchNumber	Text						Optional. Max 100 chars.
Certification									

ID	Title	XML Path	Type	N C	Lo	M e	Hi	V H	Notes
16300	Certification Name	BuildFiles/BuildFile/Certification/Certified	Text						Optional. Yellow warning if partially filled. Max 200 chars.
16310	Certification Body	BuildFiles/BuildFile/Certification/CertificationBody	Text						Optional. Max 200 chars.
16320	Certificate ID	BuildFiles/BuildFile/Certification/CertificateID	Text						Optional. Max 100 chars.
16330	Issue Date	BuildFiles/BuildFile/Certification/IssueDate	Date						Optional. Format: YYYY-MM-DD.
16340	Expiry Date	BuildFiles/BuildFile/Certification/ExpiryDate	Date						Optional. Format: YYYY-MM-DD. If any build file's certification is expired (Expiry Date < today), a red warning banner is shown on the Build Data tab. If none are expired but the package is High or Very High criticality with no build files at all, a lighter informational banner recommends adding one.
Tab-level Status Rules									
16001	Build file expiry warning	(package - level banner rule)	Requirement						If any build file's certification has expired (Expiry Date < today) → red warning banner on the Build Data tab, recommending the affected file be updated or removed. If none are expired but criticality is High or Very High and no build file has been added at all → a lighter informational banner recommends adding one; this is a recommendation, not a hard validation requirement.

TAB: Manufacturing

ID	Title	XML Path	Type	N C	Lo	M e	Hi	V H	Notes
Process Overview — General Instructions									
13010	General Instructions (Description)	ManufacturingOverview/Description	Text (long)			X	X	X	Required: Approved+ and Medium+. Multi-line.
13020	Manufacturing Method	ManufacturingOverview/ManufacturingMethod	Text			X	X	X	Required: Approved+ and Medium+. E.g. 'FDM', 'CNC Milling'. Max 200 chars.
13030	Material — top level	ManufacturingOverview/Material	Text			X	X	X	Required: Approved+ and Medium+. E.g. 'PLA', 'AlSi10Mg'. Max 200 chars.
13040	Standards	ManufacturingOverview/Standards/Standard	Text (list)						Optional. One per line. Stored as XML array.
Process Options — One per Manufacturing Route									
14001	≥1 Process Option in package	(package-level requirement)	Requirement				X	X	Package-level requirement. Required: Approved+ and High+.
14002	≥1 Process Step in option	(package-level requirement)	Requirement				X	X	Package-level requirement. Required: Approved+ and High+.
14010	Option Name	ProcessOptions/ProcessOption/OptionName	Text				X	X	Max 200 chars.
14020	Option Description	ProcessOptions/ProcessOption/Description	Text						Optional. Max 500 chars.
14030	Process Maturity	ProcessOptions/ProcessOption/Maturity	Text (select)						Optional. Values: Very Low — Ad hoc Low — Repeatable Moderate — Defined High — Measured Very High — Optimized.
14040	Option Revision	ProcessOptions/ProcessOption/Revision	Text						Optional. Option-level revision identifier e.g. 'A', '1.0'. Max 50 chars.
Implementation note: each ProcessOption additionally carries an internal id (format PO-xxxxxx), generated once when the option is created and used to name its folder under Manufacturing/ (Section 1.2). It carries no fieldId in TDP.xml; in TDP.json it appears under the synthetic key d14005 (Appendix A).									
ID	Title	XML Path	Type	N C	Lo	M e	Hi	V H	Notes
Process Steps — Basic Info (15010-15115)									
15010	Step ID	.../Steps/Step/id	Text						Automatic. Format: S-xxxxxx. Max 12 chars.
15020	Sequence Number	.../Steps/Step/sequence	Number (int)				X	X	Range: 0-9999. Ties broken by Sequence Letter, then Step ID. Displayed as 3-digit prefix in the Steps list.
15025	Sequence Letter	.../Steps/Step/sequence_letter	Text (1 char)						Optional. A-Z. Distinguishes Process Steps that share the same Sequence Number because they can be performed simultaneously (in parallel). Sorted immediately after the shared Sequence Number, before any Step ID tie-break; letter-less steps sort first.
15030	Step Name	.../Steps/Step/name	Text				X	X	Max 200 chars.
15040	Description	.../Steps/Step/description	Text (long)						Optional. Multi-line.
15050	Manufacturing Method	.../Steps/Step/manufacturing_method	Text						Optional. Max 200 chars.
15060	Build Orientation	.../Steps/Step/orientation	Text						Optional. Max 200 chars.
15070	Input Geometry	.../Steps/Step/Inputs/Input	Text (reference)						Optional. Selected from geometry file list.
15080	Output Geometry	.../Steps/Step/Outputs/Output	Text (reference)						Optional. Selected from geometry file list.
15090	Applicable Build Files	.../Steps/Step/BuildFileRefs/Ref	Text (multi-select)						Optional. Multi-select from build file list. XML array.
15100	Duration — hours	.../Steps/Step/Duration/Hours	Number (int)						Optional. Range: 0–9999.

ID	Title	XML Path	Type	N C	Lo	M e	Hi	V H	Notes
15110	Duration — minutes	.../Steps/Step/Duration/Minutes	Number (int)						Optional. Range: 0–59.
15115	Operators Required	.../Steps/Step/operator_count	Number (int)						Optional. Range: 1–99. Default: 1.
Process Steps — Materials (15200-15240)									
15200	Material	.../Steps/Step/material	Text						Optional. Max 200 chars.
15210	Material Batch	.../Steps/Step/material_batch	Text						Optional. Max 100 chars.
15220	Material Pre-treatment	.../Steps/Step/material_pretreatment	Text						Optional. Max 300 chars.
15230	Material Quantity / Consumption	.../Steps/Step/material_quantity	Text						Optional. E.g. '200 g' or '2 blanks'. Max 200 chars.
15240	Material Shelf Life / Expiry	.../Steps/Step/material_shelf_life	Text						Optional. E.g. '2025-12-31' or '12 months from opening'. Max 200 chars.
Process Steps — Environment (15300-15350)									
15300	Env Temp Min (°C)	.../Steps/Step/env_temp_min	Number (int)						Range: -100...+500. Default: 5.
15310	Env Temp Max (°C)	.../Steps/Step/env_temp_max	Number (int)						Range: -100...+500. Default: 35. Must be ≥ min.
15320	Env Humidity Min (% RH)	.../Steps/Step/env_humidity_min	Number (int)						Range: 0–100. Default: 30.
15330	Env Humidity Max (% RH)	.../Steps/Step/env_humidity_max	Number (int)						Range: 0–100. Default: 70. Must be ≥ min.
15340	Airflow / Ventilation	.../Steps/Step/airflow	Text						Optional. Max 200 chars.
15350	Cleanroom Class	.../Steps/Step/env_cleanroom_class	Text						Optional. ISO 14644 class or equivalent. E.g. 'ISO 7' / 'Class 10000'. Max 100 chars.

ID	Title	XML Path	Type	N C	Lo	M e	Hi	V H	Notes
Process Steps — Equipment (15400-15500)									
15400	Machine / Equipment	.../Steps/Step/machine	Text						Optional. Max 200 chars.
15410	Tools	.../Steps/Step/tools	Text						Optional. Max 200 chars.
15420	Software	.../Steps/Step/software	Text						Optional. Max 200 chars.
15430	Power (kW)	.../Steps/Step/power_kw	Number (float)						Optional. Range: 0–99999. 2 decimals.
15440	Compressed Air (bar)	.../Steps/Step/compressed_air_bar	Number (float)						Optional. Range: 0–99999. 2 decimals.
15450	Vacuum	.../Steps/Step/vacuum	Text						Optional. Max 100 chars.
15460	Cooling	.../Steps/Step/cooling	Text						Optional. Max 100 chars.
15470	Floor Area	.../Steps/Step/floorarea	Text						Optional. E.g. '4 × 6 m'. Max 50 chars.
15480	Equipment Height	.../Steps/Step/eqpheight	Text						Optional. E.g. '2.5 m'. Max 50 chars.
15490	Equipment Weight (kg)	.../Steps/Step/weight	Number (float)						Optional. Range: 0–999999 kg.
15500	Other Equipment Needs	.../Steps/Step/other_equipment	Text						Optional. Max 500 chars.
Process Steps — Quality (15600-15690)									
15600	Verification Methods	.../Steps/Step/InspectionMethods/Method	Text (multi-select)					X	Required: Approved+ and Very High+. Multi-select from inspection catalog. XML array.
15610	Characteristics to Verify	.../Steps/Step/inspection_points	Text						Optional. Max 500 chars.
15620	Sampling Scheme	.../Steps/Step/sampling_plan	Text						Optional. Max 500 chars.
15630	Dimensional Requirements / Tolerances	.../Steps/Step/tolerances	Text						Optional. E.g. '±0.1 mm'. Max 200 chars.
15640	Surface Texture Requirements	.../Steps/Step/surface_quality	Text						Optional. E.g. 'Ra ≤ 1.6 μm'. Max 200 chars.
15650	Conformity Criteria	.../Steps/Step/acceptance_criteria	Text (long)						Optional. Multi-line.
15660	Nonconformity Handling	.../Steps/Step/nonacceptance_handling	Text						Optional. Max 500 chars.
15670	Required Documents	.../Steps/Step/required_docs	Text						Optional. Max 500 chars.
15680	Competence Requirements	.../Steps/Step/required_training	Text						Optional. Max 500 chars.
15690	Requirement Source	.../Steps/Step/requirement_source	Text						Optional. Origin of quality requirements e.g. drawing, standard, customer spec. Max 500 chars.
Process Steps — Safety (15700-15750)									
15700	PPE	.../Steps/Step/ppe	Text						Optional. Max 300 chars.
15710	Chemical Risks	.../Steps/Step/chemical_risks	Text						Optional. Max 500 chars.
15720	Thermal Risks	.../Steps/Step/thermal_risks	Text						Optional. Max 500 chars.
15730	Mechanical Risks	.../Steps/Step/mechanical_risks	Text						Optional. Max 500 chars.
15740	ESD Risks	.../Steps/Step/esd_risks	Text						Optional. Max 500 chars.
15750	Other Safety Notes	.../Steps/Step/other_safety	Text						Optional. Max 500 chars.

ID	Title	XML Path	Type	N C	Lo	M e	Hi	V H	Notes
Process Steps — Post-processing & Parameters (15800-15910)									
15800	Post-processing	.../Steps/Step/post_processing	Text (long)						Optional. Multi-line.
15810	Notes	.../Steps/Step/notes	Text (long)						Optional. Multi-line.
15900	Parameter name	.../Steps/Step/Parameters/Parameter/@name	Text						Optional. Key-value pair. Max 100 chars. XML attribute: @name.
15910	Parameter value	.../Steps/Step/Parameters/Parameter	Text						Optional. Always paired with 15900. Max 500 chars.
Process Steps — Attachments (15950-15990)									
15950	Attachment file name	.../Steps/Step/Attachments/Attachment/FileName	Text						Automatic. Read-only.
15960	Attachment source path	.../Steps/Step/Attachments/Attachment/SourcePath	Text						Automatic. Read-only.
15970	Attachment format	.../Steps/Step/Attachments/Attachment/FileFormat	Text						Automatic. Read-only.
15980	Attachment size (bytes)	.../Steps/Step/Attachments/Attachment/FileSize	Text (number)						Automatic. Read-only.
15990	Attachment checksum (SHA-256)	.../Steps/Step/Attachments/Attachment/Checksum	Text (hash)						Automatic. 64 hex chars. Read-only.

TAB: Logistics and Safety

ID	Title	XML Path	Type	N C	Lo	M e	Hi	V H	Notes
Packing Requirements									
19010	Packing Material	PackingAndSafety/Pack ingMaterial	Text (select)						Optional. Values: Paper and cardboard Plastics Metals Glass Wood and wood-based Ceramics Textiles and fibre.
19020	Cushioning	PackingAndSafety/Cus hioning	Text						Optional. Max 200 chars.
19030	Moisture Protection	PackingAndSafety/Moi stureProtection	Text						Optional. Max 200 chars.
19040	Surface Protection	PackingAndSafety/Sur faceProtection	Text						Optional. Max 200 chars.
19050	Label — Fragile	PackingAndSafety/Lab elFragile	Boolean						Optional. true false.
19060	Label — This Side Up	PackingAndSafety/Lab elThisSideUp	Boolean						Optional. true false.
19070	Label — Do Not Stack	PackingAndSafety/Lab elDoNotStack	Boolean						Optional. true false.
19080	Label — ESD	PackingAndSafety/Lab elESD	Boolean						Optional. true false.
19090	Other Labels	PackingAndSafety/Oth erLabels	Text						Optional. Max 200 chars.
Handling Instructions									
19100	Allowed Lifting Methods	PackingAndSafety/All owedLifting	Text (long)						Optional. Multi-line. Max 500 chars.
19110	Forbidden Lifting Methods	PackingAndSafety/For biddenLifting	Text (long)						Optional. Multi-line. Max 500 chars.
19120	Centre of Gravity Description	PackingAndSafety/COG Description	Text						Optional. Max 300 chars.
19130	Max Handling Forces	PackingAndSafety/Han dlingForces	Text						Optional. Max 200 chars.
19140	ESD Requirements — handling	PackingAndSafety/ESD Requirements	Text						Optional. Max 200 chars.

ID	Title	XML Path	Type	N C	Lo	M e	Hi	V H	Notes
Transport Requirements									
19200	Transport Orientation	PackingAndSafety/TransportOrientation	Text						Optional. Max 200 chars.
19210	Securing Method	PackingAndSafety/SecuringMethod	Text						Optional. Max 200 chars.
19220	Special Transport	PackingAndSafety/SpecialTransport	Text						Optional. Max 200 chars.
19230	HAZMAT Classes (repeating)	PackingAndSafety/HazmatClasses/Class	Text (list)						Optional. XML array. E.g. 'UN 1263 Class 3'.
19240	Transport Temp Min (°C)	PackingAndSafety/TempMin	Number (float)						Optional. Range: -100...+9999. 1 decimal. Default: -20.
19250	Transport Temp Max (°C)	PackingAndSafety/TempMax	Number (float)						Optional. Range: -100...+9999. Must be ≥ min. Default: 60.
19260	Transport Humidity Min (% RH)	PackingAndSafety/HumidityMin	Number (float)						Optional. Range: 0–100. Default: 10.
19270	Transport Humidity Max (% RH)	PackingAndSafety/HumidityMax	Number (float)						Optional. Range: 0–100. Must be ≥ min. Default: 90.
19280	Stackable	PackingAndSafety/Stackable	Boolean						Optional. true false.
19290	Max Stack Layers	PackingAndSafety/MaxStackLayers	Number (int)						Optional. Range: 0–100.
Safety Information									
19300	PPE — packing	PackingAndSafety/PPE	Text						Optional. Max 300 chars.
19310	Edge Risks	PackingAndSafety/EdgeRisks	Text						Optional. Max 300 chars.
19320	Item Weight (kg)	PackingAndSafety/Weight	Number (float)						Optional. Range: 0–999999 kg.
19330	Manual Lift Limit (kg)	PackingAndSafety/ManualLiftLimit	Number (float)						Optional. Range: 0–9999 kg.
19340	Chemical Risks — packing	PackingAndSafety/ChemicalRisks	Text (long)						Optional. Multi-line.
19350	Thermal Risks — packing	PackingAndSafety/ThermalRisks	Text (long)						Optional. Multi-line.
19360	Other Safety Notes — packing	PackingAndSafety/OtherSafety	Text (long)						Optional. Multi-line.

TAB: Additional Data

ID	Title	XML Path	Type	N C	Lo	M e	Hi	V H	Notes
Additional CAD / Reference Models — Repeating Record									
17010	File Name	OptionalData/AdditionalCAD/CADFile/FileName	Text						Automatic. Supported: STL STEP STP IGES 3MF. Max 260 chars.
17020	File Format	OptionalData/AdditionalCAD/CADFile/FileFormat	Text						Automatic. From extension. Read-only.
17030	File Size (bytes)	OptionalData/AdditionalCAD/CADFile/FileSize	Text (number)						Automatic. Read-only.
17040	SHA-256 Checksum	OptionalData/AdditionalCAD/CADFile/Checksum	Text (hash)						Automatic. 64 hex chars. Read-only.
17050	Description	OptionalData/AdditionalCAD/CADFile/Description	Text						Optional. E.g. 'Post-build scan', 'Reference assembly'. Max 500 chars.
17060	Source Path	OptionalData/AdditionalCAD/CADFile/SourcePath	Text						Automatic. Local path. Read-only.
Reference Images — Repeating Record									
17100	File Name	OptionalData/Images/Image/FileName	Text						Automatic. Supported: PNG JPG JPEG BMP TIFF. Max 260 chars.
17110	File Format	OptionalData/Images/Image/FileFormat	Text						Automatic. Read-only.
17120	File Size (bytes)	OptionalData/Images/Image/FileSize	Text (number)						Automatic. Read-only.
17130	SHA-256 Checksum	OptionalData/Images/Image/Checksum	Text (hash)						Automatic. 64 hex chars. Read-only.
17140	Description	OptionalData/Images/Image/Description	Text						Optional. Max 500 chars.
17150	Source Path	OptionalData/Images/Image/SourcePath	Text						Automatic. Local path. Read-only.

5. Archive Layout Reference

A worked example of a saved .tdp archive's directory listing, showing the Manufacturing/ attachment layout and the History/ audit trail described in Sections 1.2, 1.4 and 1.5:

example.tdp (ZIP archive)

```
|-- TDP.xml
|-- TDP.json
|-- Geometry/
|   |-- bracket.stl
|-- BuildFiles/
|   |-- job.gcode
|-- Optional/
|   |-- CAD/
|   |   |-- reference.step
|   |-- Images/
|   |   |-- inspection_photo_ref.png
|-- Manufacturing/
|   |-- PO-a1b2c3/                (Process Option internal id)
|   |   |-- S-000123/            (Process Step id)
|   |   |-- weld_photo.jpg
|-- History/
|   |-- CH-0018026E.txt           (first save)
|   |-- CH-BA4CF889.txt           (second save, same checksum: no
|   |                               real content changed)
|   |-- CH-210BF43D.txt           (third save, checksum changed)
```

A Process Option with two steps that can be carried out at the same time would be numbered, for example, Sequence Number 2 with Sequence Letters A and B (field 15025); both steps' folders still live under Manufacturing/<OptionID>/, one per Step ID, regardless of shared sequence numbering.

checksum_debug.txt (not shown above) is written next to example.tdp on disk, not inside the archive — see Section 1.2.

Appendix A: TDP.json Non-Spec Identifiers

The following containers and values carry no fieldId in TDP.xml. TDP.json therefore uses the synthetic (non-spec) numeric keys below purely as stable identifiers — they do not appear in Section 4 and must never be interpreted as spec-numbered fields.

Container / Field	Synthetic Key	Meaning
Version/ChangeHistory	d11090	Array of Change History entries
ChangeHistory/ChangeEntry/Checksum	d11095	Per-save audit checksum (Section 1.5); excluded from the package checksum it records
Geometry	d12000	Array of Geometry files
ProcessOptions	d14000	Array of Process Options
ProcessOption/id	d14005	Process Option's internal identifier (Manufacturing/ folder name); not a numbered field in TDP.xml
ProcessOption/Steps	d15005	Array of Process Steps within one option
Step/Attachments	d15940	Array of step-level attachments
Step/Parameters	d15890	Array of {"d15900": name, "d15910": value} objects
BuildFiles	d16000	Array of Build Files
OptionalData/AdditionalCAD	d17000	Array of Additional CAD files
OptionalData/Images	d17090	Array of Reference Images

Standards (ManufacturingOverview/Standards) and HAZMAT Classes (PackingAndSafety/HazmatClasses) are not listed above: their repeated text values (Standard, Class) already carry the real fieldId of the field they repeat (13040 and 19230 respectively), so no synthetic key is needed for them.